ReDI School of Digital Integration

Machine Learning Final Project

Autumn 2026

We use tech to connect human potential and opportunity with dignity & humility

Important Dates

Sunday, Nov 29 (by 23:59)	Submit project + presentation on Google Classroom
Tuesday, Dec 1	Presentations (online)
Thursday, Dec 3	Presentations (online)
Tuesday, Dec 8 (by 23:59)	Teacher deadline to submit final grades
Wednesday, Dec 9	Demo Day

Project Requirements

Language: Python (with scikit-learn, pandas, numpy, matplotlib, seaborn, …)

Requirements

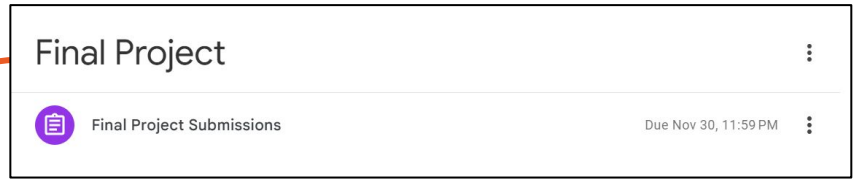
1. **Problem Statement and Dataset Selection**
   - Define problem statement and objectives
   - Select dataset and justify the chosen ML technique
1. **Data Preparation and Exploration**
   - Perform data cleaning and preprocessing
   - Conduct exploratory data analysis and visualization
1. **Model development and evaluation**
   - Implement the chosen ML technique (tune model parameters as necessary)
   - Evaluate the model using appropriate metrics
   - Interpret the results
   - (optional) Compare performance of different models
1. **Ethical Considerations and Limitations**
   - Evaluate model fairness + limitations
   - Propose strategies to mitigate bias (if applicable)
1. **Conclusion and Future Work**
   - Develop a coherent narrative that ties your findings to the original problem statement.
   - Summarize key findings and their implications.
   - Suggest potential areas for future analysis or data collection

The focus is not just on achieving the "right" result but on demonstrating a clear, well-reasoned process that shows understanding of the data science workflow.

Presentation Requirements

Your presentation:

- must be no longer than 5 minutes
- will be followed by ~10 minutes questions and feedback from your teachers/fellow students
- must use this ReDI presentation template
- must be submitted on Google Classroom **by Nov 29 @ 23:59**

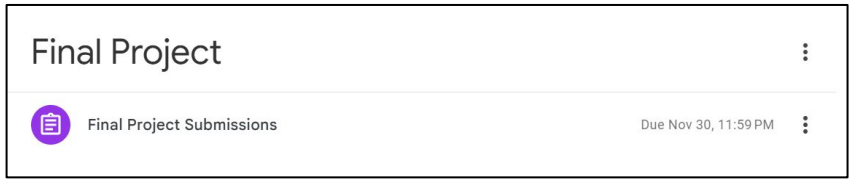


What to include in your presentation:

- Why you chose this project/idea/dataset?
- What did you do? What was the process you followed?
- Why did you choose a particular modeling technique?
- What is the result of your project? What is the impact? What are the real world implications?
- Did you face any difficulties or roadblocks? How did you overcome them?
- Conclusions: What are some future improvements? How could the project be used?

Submission Requirements

On Google Classroom, submit (**by Nov 29 at 23:59**):



1. all project files in a compressed folder (ZIP file) including:
    - Jupyter Notebook with all code, analysis, and documentation
    - Original and cleaned datasets or link to online original data set
    - Presentation slides (PPT or PDF format)
    - Any additional visualizations or documents created
1. a README text file with:
    - Project overview
    - List of included files
    - Instructions to run the analysis (including required Python libraries)

Optional: If familiar with GitHub, you may submit your project as a GitHub repository link instead of a ZIP file.

Timeline + Milestones

Machine Learning Project Timeline

- **Week 1 (Oct 12 - Oct 18):** Dataset selection, and initial analysis.
- **Week 2 (Oct 19 - Oct 25):** Data cleaning, preprocessing, and feature engineering.
- **Week 3 (Oct 26 - Nov 1):** Model selection, initial training, and evaluation.
- **Week 4 (Nov 2 - Nov 8):** Model tuning, final evaluation, and comparison.
- Career week (Nov 9 - Nov 15)
- **Week 5 (Nov 16 - Nov 22):** Report writing, visualisations, and **final submission**.
- **Week 6 (Nov 23 - Nov 29):** Project check-ins

Key Deliverables: Project proposal, data preprocessing report, initial model evaluation, final model report, and final submission on **November 29.**

Weekly Check-ins recommended for guidance and feedback.

Project Ideas

(but feel free to come up with your own!)

Some Ideas

Customer Churn Prediction

- Predict which customers are likely to leave a service

Credit Risk Assessment

- Evaluate the creditworthiness of loan applicants

House Price Prediction

- Estimate house prices based on various features

Disease Diagnosis

- Classify medical conditions based on patient data

Spam Email Detection

- Identify spam emails using text classification

Stock Price Prediction

- Forecast stock prices using historical data

Customer Segmentation

- Group customers based on purchasing behavior

Fraud Detection

- Identify fraudulent transactions in financial data

Some Data Sources



UCI Repository

Kaggle Datasets

World Bank Open Data

OECD Data (Organisation for Economic Co-operation and Development)

Harvard Dataverse

Kdnuggets

FreeCodeCamp

Grading + Certificates

Grading

You will be graded on several elements, including:

- Problem Formulation: Clarity and relevance of the problem statement
- Data Preprocessing: Effectiveness of data cleaning and feature engineering
- Model Selection: Appropriateness of chosen algorithms for the problem
- Implementation: Correct implementation of ML algorithms and techniques
- Model Evaluation: Thorough evaluation using appropriate metrics
- Interpretation: Depth of model interpretation and insights derived
- Code Quality: Readability, efficiency, and documentation of code
- Presentation: Clarity and engagement of the final presentation
- Documentation: Comprehensiveness and clarity of project documentation
- AI Use: Appropriateness, transparency and criticality of AI tools use

Regardless of the grade you receive, teachers will provide constructive feedback for you to improve!

Certification & Completing the Course

Certificate of Achievement with Honours

→ Indicates exceptional performance and mastery of the subject matter

a. Completes an exceptional final project i.e. beyond expected requirements
b. Completes career services requirements
c. Attends 80% of classes

Certificate of Achievement

→ Indicates successful completion and competence in the subject matter

b. Completes a final project that meets minimum requirements
c. Completes career services requirements
d. Attends 80% of classes

Certificate of Participation

→ Acknowledges engagement in the course

a. Does not complete final project / fails final project i.e. does not meet minimum requirements
b. Completes career services requirements
c. Attends 80% of classes

No certificate

→ Student fails to meet one of more of the requirements for Certificate of Participation (a-c)

Certification is determined by the DCP team and takes into account all requirements and individual circumstances.

Academic Honesty

Academic Honesty - Plagiarism

Plagiarism is the act of presenting someone else's work, ideas, or expressions as your own without proper attribution. This includes, but is not limited to:

- Copying and pasting code, text, or design elements without citation
- Paraphrasing someone else's work without acknowledgment
- Submitting work completed by another person as your own

Plagiarism is strictly prohibited at ReDI. All work submitted must be your own original creation or properly cited if using external sources. Failure to comply with this policy may result in disciplinary action, including failure of the assignment, project, or course.

Academic Honesty - AI Tools

AI tools (such as ChatGPT, GitHub Copilot, or similar) can be valuable learning aids when used responsibly. However, in order to truly benefit from these tools and develop genuine skills, keep in mind:

👍**Allowed Uses**

- Brainstorming ideas
- Debugging code,
- Summarising long texts so they are easier to digest and understand
- Learning about new concepts or technologies
- Generating example code snippets for learning purposes

👎**Prohibited Uses**

- Generating entire (or substantial portions) of your project
- Submitting AI-generated work as your own without significant modification or additional input

When using AI tools or any external resources, make sure to:

- Mention the use of AI tools in your project documentation
- Cite any resources, tutorials, or references used

Violation of this policy may result in failure of the project or course.

If you are unsure about the proper use of AI tools or have questions about plagiarism, please consult with your teachers or the DCP team. We are here to help you learn and succeed!

Demo Day

What Is Demo Day?

Demo Day - Project Fair

Pitch your project and get constructive feedback from different perspectives.

This is different from your final project presentation. There, you will explain your full process and go into detail.

Here, the **goal is to communicate your idea in a simple, engaging way** so anyone can understand it.

This is your chance to:

- Practice pitching your idea (not presenting slides) 💬
- Explain your project to non-technical people 🧠
- Receive diverse feedback & perspectives 🙋🏾‍♀️🙋🏽‍♂️🙋🏻‍♀️🙋🏼‍♀️
- Build confidence in real-world conversations 😎
- Learn how to communicate value, not just features 🚀

This is a learning experience so there is no doing “good” or “bad”, everyone is encouraged to participate and gain some valuable experience and feedback for your work!



How Does Demo Day Work?

1. Students will set up around campus to present their final projects. Everyone else (teachers, volunteers, and classmates) will walk around, explore, listen, and cheer you on.
2. You'll give a short, casual pitch of your project (3-5 minutes)

   ! The people in your group might not have the technical expertise in your field so it's best to keep your pitch about the main message and implications of what you did.

   ! You might have to eliminate some parts of the presentation you used for your class.

1. Quick Q&A and feedback from your audience — this isn't graded! It's a chance for everyone to learn about what you did.
2. Continue until all students have presented.

ReDI School of Digital Integration

Machine Learning Final Project — Do’s and Don’ts in the Use of AI

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General Policy regarding the Use of AI

- AI tools are allowed and encouraged throughout the project
- You must understand and be able to explain every part of your work
- Document where and how you used AI assistance
- Your thinking, decisions, and interpretations must be your own
- Grading: "*appropriateness, transparency and criticality of AI tools use*"

"AI is your assistant, not your replacement.
Use it to learn faster — not to skip learning."

1. Tools & Assets

Tools & Assets

✔ Use coding assistants (Copilot, ChatGPT) for syntax help, boilerplate, and debugging error messages

✔ Use data tools (pandas-profiling, sweetviz) for quick EDA overviews of your dataset

✔ Ask AI to explain unfamiliar library functions — e.g. "Explain sklearn's RandomForestClassifier parameters"

✔ Use AI to help write repetitive data-cleaning code (handling missing values, encoding categorical features)

✘ Don't let AI choose your dataset or define your problem — that decision is yours

✘ Don't use AI-generated code you cannot explain line by line

✘ Don't use AI to fabricate data or generate synthetic results you present as real

✘ Don't rely on a single tool — cross-reference suggestions from different sources

Discussion

Which AI tools have you already tried?
Share one thing that worked well and one thing that didn't.

2. Process & Workflow

Process & Workflow

✔ Use AI to brainstorm approaches — e.g. "What ML models work well for classification with imbalanced data?"

✔ Use iterative prompting: prompt → review → refine → verify

✔ Use AI to draft outlines for your project report structure

✔ Use AI to help you get unstuck — describe the problem, paste the error, ask for guidance

✘ Don't ask AI to "build my entire project" — you learn nothing and can't defend your choices

✘ Don't skip understanding the data by just feeding it to AI

✘ Don't accept the first AI suggestion without exploring alternatives

✘ Don't let AI make judgment calls: which features matter, which model to choose — that requires YOUR reasoning

Activity: Workflow Walkthrough

Scenario: You are stuck on a data-cleaning step — your DataFrame has mixed date formats and missing values.

How would you use AI to help?

→ What prompt would you type into ChatGPT / Copilot

→ How would you verify the AI's suggestion is correct?

3. Quality and Evaluation

Quality & Evaluation

✔ Always run AI-generated code and check the output against expected results

✔ Verify statistics or metrics AI produces by computing them yourself (or with a second method)

✔ Ask AI to explain its reasoning — if the explanation doesn't make sense, don't trust the code

✔ Use AI to help you interpret model evaluation metrics (precision, recall, F1, ROC-AUC)

✘ Don't trust AI-generated data analysis without verification — AI can hallucinate patterns that don't exist

✘ Don't copy AI-generated visualisation code without understanding what it plots and why

✘ Don't accept "This model is good" from AI — check metrics, check for overfitting, validate on test data

✘ Don't ignore warnings or errors that AI tells you to suppress

Activity: Spot the Problem

An AI assistant says:
"Your model achieved 98 % accuracy, so it is ready for deployment."

What questions should you ask before trusting this claim?

→ Hint: Think about class imbalance, overfitting, test vs. train split, and appropriate metrics

4. Ethics and Governance

Ethics & Governance

✔ Be transparent — document where and how you used AI in your project

✔ Check your dataset for bias before training (remember the HR case study!)

✔ Consider privacy: do not upload personal or sensitive data to public AI tools

✔ Think about who your model could affect and how — discuss limitations honestly

✘ Don't present AI-generated text or code as entirely your own work without acknowledgement

✘ Don't ignore bias in your data or model — acknowledge and discuss limitations

✘ Don't use AI to generate conclusions you haven't verified against the data

✘ Don't scrape or use datasets without checking their licence and terms of use

Discussion: Data Ethics Scenario

Your dataset contains names, ages, and nationalities.

What do you need to think about before:
a) uploading it to ChatGPT for analysis?
b) training a model that predicts hiring decisions?

→ Consider: privacy, consent, bias, fairness, GDPR

→ Discuss in small groups for 3 minutes

→ Each group shares one key concern

5. Learning & Skills

Learning & Skills

✔ Use AI to learn faster — ask it to explain concepts, compare approaches, and give examples

✔ Focus on skills AI cannot replace: problem framing, critical thinking, domain understanding, communication

✔ Practice writing effective prompts — it is a real professional skill

✔ Learn to evaluate AI output — this IS the new core skill

✘ Don't use AI as a shortcut that skips understanding — the exam is in your head, not on screen

✘ Don't stop practising fundamentals (Python, statistics, data wrangling) because "AI can do it"

✘ Don't mistake AI fluency for your own understanding — can you explain it without AI?

✘ Don't over-rely: if the AI tool goes down, you should still be able to work

Quick Reference

Quick Reference

DO

✔ Use AI for syntax, debugging, and explanations

✔ Iterate: prompt → review → refine → verify

✔ Always test and verify AI outputs

✔ Be transparent about AI use

✔ Focus on understanding, not just output

DON'T

✘ Don't use code you can't explain

✘ Don't let AI make your key decisions

✘ Don't trust AI statistics without checking

✘ Don't present AI work as entirely your own

✘ Don't skip learning fundamentals

Before you go…

Please take 2 minutes to give us feedback on today's lesson.

This feedback is **anonymous** and only seen by **the DCP team**!



Scan the QR code or go to: